IT462: Exploratory Data Analysis

Lab Assignment: Data Cleaning, Outliers, Missing Values and Feature Engineering

Dataset

Use **one** of the following (state which one in your report). Both are freely available and have numeric, categorical and missing data:

- **Option A (recommended):** Titanic (`seaborn.load_dataset('titanic')`). Target: `survived`.
- **Option B:** California Housing (`sklearn.datasets.fetch_california_housing(as_frame=True)`). Target: `MedHouseVal`. This one has no missing values, so first create about 10% missing values at random in 3 numeric columns using `np.random.seed(42)`.

Submission rules

- Submit one Jupyter notebook (`.ipynb`) with code, outputs and markdown answers, plus a PDF export of it.
- Use `random_state=42` everywhere so results are reproducible.
- Every plot needs a title and labelled axes. Every answer needs a 1 to 3 line interpretation, not just output.
- File name: `IT462_Lab_<RollNo>.ipynb`

Questions A: Understanding the Dataset

Before cleaning anything, know what you have.

Q1. Load the dataset. Print its **shape**, column names, data types (`df.info()`) and `df.describe(include='all')`.

Q2. Classify every column as *numerical (continuous / discrete)* or *categorical (nominal / ordinal)* in a table. Mark the target column and any ID-like column that should not be used as a feature.

Q3. Check for duplicate rows and for columns with a single constant value. Remove them if found and state how many you removed.

Questions B: Outlier Detection and Treatment

An **outlier** is an observation that lies far from the bulk of the data. Detection is done at three levels:

Level	Looks at	Tools
Univariate	One variable at a time	Boxplot, histogram, Z-score, IQR rule
Bivariate	Two variables together	Scatter plot, pair plot
Multivariate	Many variables together	Mahalanobis distance, Isolation Forest / LOF

B1. Univariate

Q4. Pick **three** numeric columns. For each, draw a **histogram** and a **boxplot** side by side. Comment on skewness and visible outliers.

Q5. Compute outliers using both rules and report the count for each column:

IQR rule: outlier if x < Q1 - 1.5 x IQR or x > Q3 + 1.5 x IQR

Z-score rule: outlier if |z| > 3

Why do the two rules disagree on skewed columns? (2 to 3 lines)

```
def iqr_outliers(s):
    q1, q3 = s.quantile([.25, .75]); iqr = q3 - q1
    return s[(s < q1 - 1.5*iqr) | (s > q3 + 1.5*iqr)]
```

B2. Bivariate

Q6. Draw a **scatter plot** of two numeric variables that you expect to be related (for example fare vs age, or MedInc vs MedHouseVal), then a **pair plot** of 4 numeric columns. Circle (or list) one point that looks normal in each variable alone but unusual in combination.

B3. Multivariate

Q7. Compute the **Mahalanobis distance** of each row (numeric columns, no missing values) and flag rows above the 97.5th percentile of the chi-square distribution with *p* degrees of freedom. Report how many rows are flagged. Optionally compare with `IsolationForest(contamination=0.02)`.

```
from scipy.spatial.distance import mahalanobis
X = df[num_cols].dropna()
mu, VI = X.mean().values, np.linalg.inv(np.cov(X.values, rowvar=False))
d2 = X.apply(lambda r: mahalanobis(r.values, mu, VI)**2, axis=1)
cutoff = stats.chi2.ppf(0.975, df=X.shape[1])
flagged = X[d2 > cutoff]
```

B4. Treatment

Q8. Apply **three** treatments to one skewed column: (a) **capping** at the IQR fences, (b) **Winsorization** at 5% on each tail, (c) **log transform** (`np.log1p`). Plot before/after boxplots for all three.

Q9. Make a table of mean, median, standard deviation and skewness for the original and each treated version. Which treatment would you choose and why? When would you **not** remove an outlier?

```
cap = s.clip(lower=q1 - 1.5*iqr, upper=q3 + 1.5*iqr)
win = pd.Series(winsorize(s, limits=[0.05, 0.05]), index=s.index)
logt = np.log1p(s)
```

Questions C: Missing Values

C1. Detection

Q10. Report the count and percentage of missing values per column as a sorted table. Which columns exceed 30% missing?

Q11. Using `missingno`, draw: (a) a **bar chart** (`msno.bar`), (b) a **matrix** (`msno.matrix`), (c) a **heatmap** (`msno.heatmap`) and (d) a **dendrogram** (`msno.dendrogram`). Explain what each one tells you.

Q12. Classify the missingness of the column with most missing data as **MCAR**, **MAR** or **MNAR** and justify with evidence (for example compare the mean of another variable between rows where the column is missing and not missing; use a t-test or a bar plot).

Type	Meaning
MCAR	Missing completely at random: missingness is unrelated to any variable.
MAR	Missing at random: depends on other *observed* variables only.
MNAR	Missing not at random: depends on the missing value itself.

C2. Treatment

Q13. Apply the following to the numeric columns, each on a fresh copy of the data: (a) **deletion** (listwise), (b) **mean** imputation, (c) **median** imputation, (d) **mode** imputation for categorical columns (`SimpleImputer`). Report rows kept and the effect on mean and standard deviation of the imputed column.

Q14. Apply **MICE** (`IterativeImputer`, `max_iter=10`, `sample_posterior=True`). Produce **three** imputed datasets using different `random_state` values and compare the imputed values of the same cells.

Q15. Plot the **density (KDE)** of one imputed column for: original (observed only), mean-imputed, median-imputed and MICE-imputed. Which method distorts the distribution most and why? Which preserves relationships between variables best?

```
imp = IterativeImputer(max_iter=10, sample_posterior=True, random_state=0)
X_mice = pd.DataFrame(imp.fit_transform(df[num_cols]), columns=num_cols, index=df.index)
```

Questions D: Feature Engineering, Selection and Sensitivity

D1. Encoding and scaling

Q16. Encode categorical variables: **label / ordinal** encoding for ordered categories and **one-hot** encoding for nominal ones. Explain why label-encoding a nominal feature is risky for linear models.

Q17. Create at least **three new features** (examples: family size = sibsp + parch + 1, log of a skewed variable, ratio or interaction of two columns, bins of age). Justify each in one line. Then apply **StandardScaler** and **MinMaxScaler** and state which models need scaling.

D2. Feature selection

Apply one method from each family and compare the selected features:

Family	Idea	Use in this lab
Filter	Score features independently of the model	Correlation, chi-square / mutual information (`SelectKBest`)
Wrapper	Search subsets using model performance	RFE with Logistic / Linear Regression
Embedded	Selection happens during training	Lasso (L1) or Random Forest importance

Q18. Implement the three methods (top 5 features each) and display the results in one table. Which features are selected by all three?

Q19. Plot the correlation heatmap of the final numeric features and remove one feature from every pair with |r| > 0.85. Explain how multicollinearity affects model interpretation.

D3. Sensitivity analysis

Sensitivity analysis asks: *do my conclusions change when I change my cleaning choices?*

Q20. Train the **same model** (Random Forest, `random_state=42`, 5-fold cross-validation) on the data produced by each pipeline below and report mean and std of the CV score (accuracy for classification, R^2 for regression):

Pipeline	Outliers	Missing values
P1	None treated	Mean / mode imputation
P2	Capping (IQR)	Median / mode imputation
P3	Winsorization 5%	MICE
P4	Log transform	MICE

Q21. Present the four scores as a bar chart with error bars. Write a short conclusion (5 lines max): which pipeline is best, is the difference larger than the CV standard deviation, and what would you recommend?

Common mistakes to avoid

- Fitting imputers or scalers on the full dataset before the train/test split (data leakage). In Q20, put them inside a `Pipeline` so they are fitted only on training folds.
- Deleting outliers without checking whether they are valid, rare observations.
- Applying MICE to the target column or including the target when evaluating the model fairly.
- Reporting results without any interpretation.